ObjectWidth [SimTalk] - Button

Sets the **Width** of the *Button* designated by <Path> with which it is shown in the *Frame*.

Remarks

If you enter a long label, you have to adjust the width so that the label fits on the button without the text being cut off.

Type

Attribute

Syntax

```
<Path>.ObjectWidth:integer
```

Assignment Value

You can assign a value of data type *integer*.

Example

```
MyButton.ObjectWidth := 4 // meters
```

SimTalk

Width [text box] - Button

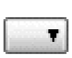
ObjectHeight [SimTalk] - Button

DropDownList

DropDownList [object]

Use the object *DropDownList* for showing a drop-down list in the *Frame*. When you select one of the items, it executes the action which you programmed in the control.

Image

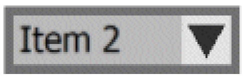
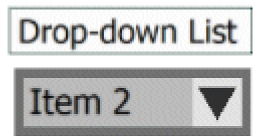
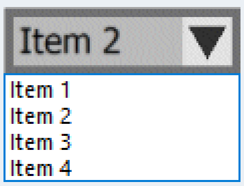
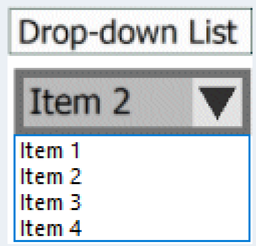


Description

The *DropDownList* provides the user with several options, one of which he can select. You can define the items, which the *DropDownList* shows, either in its dialog or with the attribute **Items**.

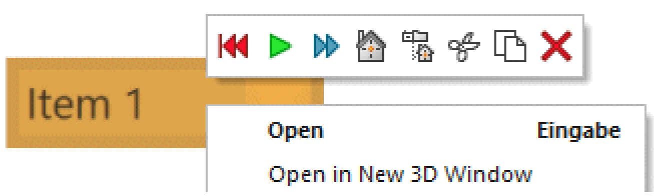
Whenever the user of your *DropDownList* selects another item, the drop-down list executes the action which you programmed in the **Control**.

The *DropDownList* looks like this by default:

Appearance without caption	Appearance with caption
	
	

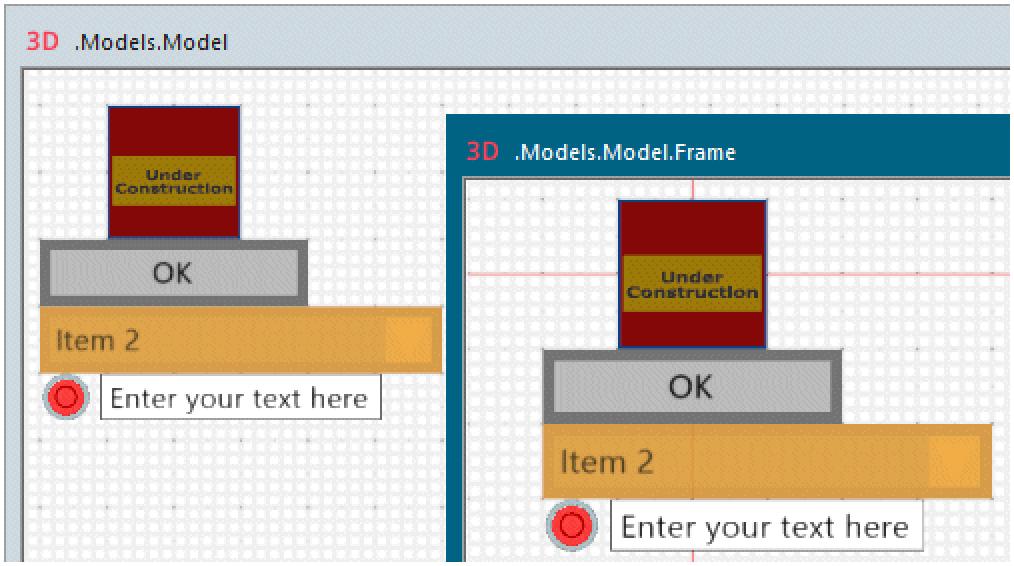
> **Note**
> *Plant Simulation* only accepts the click the *DropDownList* if it occupies at least 10 pixels in each direction in the model window. If you zoom out very far, the picture of the *DropDownList* might become too small, preventing *Plant Simulation* to recognize it reliably.

To open its dialog box, click the drop-down list with the right mouse button and select **Open** on the context menu.

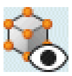


To select the drop-down list in the *Frame* window, you can also drag a marquee over its icon.

You can also click a *DropDownList* and select an item in it in a *sub-Frame* and trigger the respective action.



To show a tooltip with information about the *DropDownList*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *DropDownList*, click **Show Manipulators**  on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model

To add the object *DropDownList* to your simulation model, click  **Manage Class Library** > **Basic Objects > UserInterface > DropDownList** on the **Home** ribbon tab.

See also

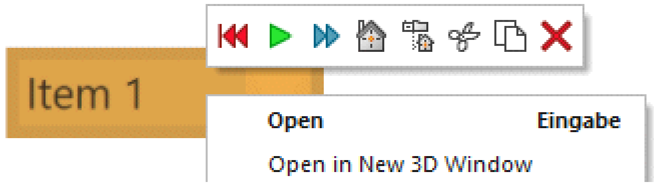
Select an Option from a Drop-down List

Configure the Check Box and the Drop-down List

Dialog Box of the DropDownList

Dialog Box of the DropDownList

Click the *DropDownList*, which you inserted into a *Frame*, with the right mouse button and select **Open** on the context menu to open its dialog box.

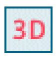


Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under **Dialog Items of the Objects**.

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box **Edit 3D Properties**:

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the **Tab Attributes**.

Width [text box] - DropDownList

Type in the **Height** in meters with which the *DropDownList* is shown in the *Frame*.

Remarks

If you type in long names for the **Items**, you have to adjust the width so that the names of the items in the drop-down list are not cut off.

To change the width, you can also hold down **Ctrl+Shift** and drag the left or right side of the icon. *Plant Simulation* shows the width in meters.

SimTalk

ObjectWidth [SimTalk] - DropDownList

Height [text box] - DropDownList

Type in the **Height** in meters with which the *DropDownList* is shown in the *Frame*.

Remarks

If you type in a **Height** of more than 30 pixels, *Plant Simulation* uses a larger font to display the **Label** of the drop-down list.

To change the height, you can also hold down **Ctrl+Shift** and drag the top or the bottom of the icon.

SimTalk

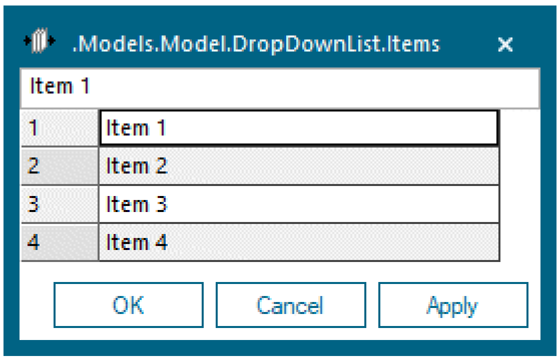
ObjectHeight [SimTalk] - DropDownList

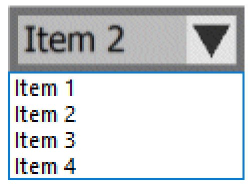
Value [text box] - DropDownList

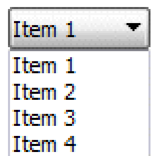
Type in which item in the *DropDownList* will be selected when you open it. The value is identified by its index number.

Remarks

This is one of the items which you typed into the list of **Items**.







Note

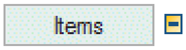
If you localize the items to another language, you will usually use the **Value** instead of the attribute **Item**. The value is a number which is language-independent, while the *Item* is a string which differs from language to language.

SimTalk

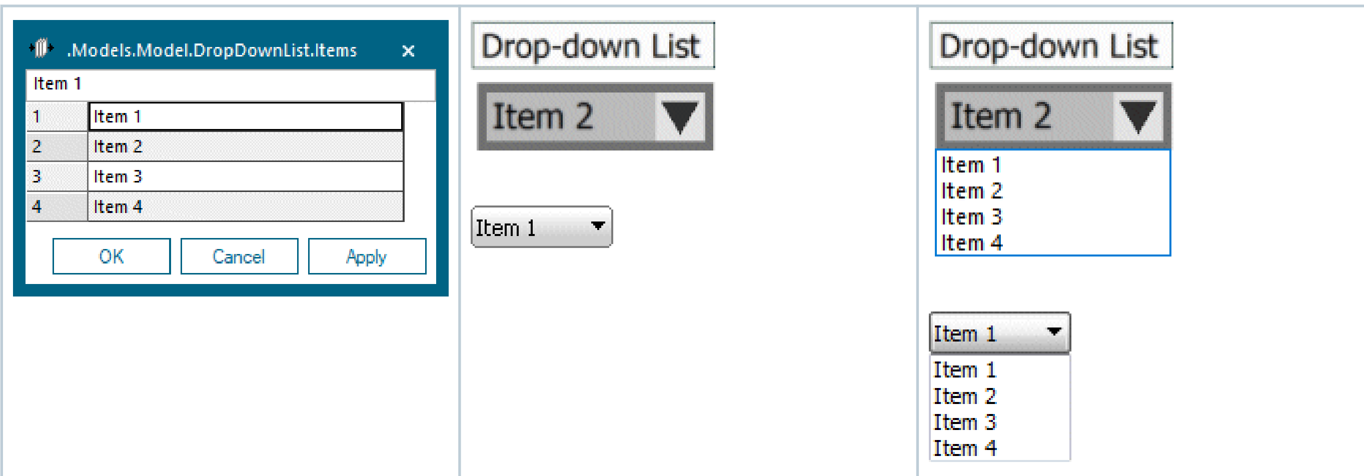
Value [SimTalk] - DropDownList

Items [button] - DropDownList

To open the list, into which you type the **Items** which the *DropDownList* displays when you click it, click this button.



Remarks



If you want to use the **index number** to access an item in the list, use the attribute **Value**.

If you want to use the **name** to access an item in the list, use the attribute **Item**.

SimTalk

Items [SimTalk] - DropDownList

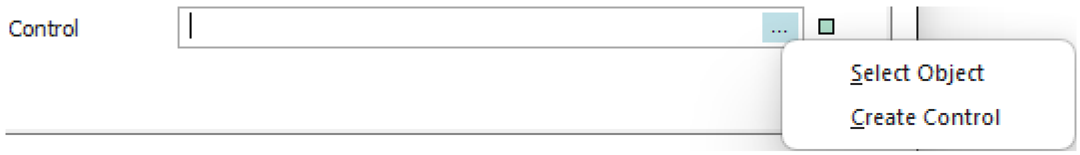
Item [SimTalk] - DropDownList

See also

Configure the Feeder Line with Sensor and Sensor Control

Control [DropDownList]

Modifies the built-in behavior of the object. The object calls the **Control** when you select an item from the *DropDownList* in the *Frame*.



Remarks

The anonymous identifiers **?** and **@** in the **Control** are set to the *DropDownList* when the method is called.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

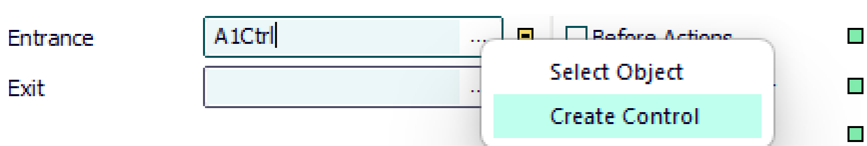
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

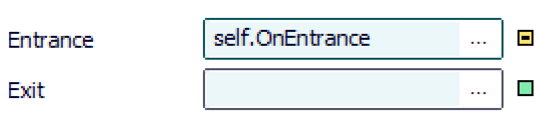
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.

- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

A **control** might, for example, look like this:

```
switch ?.Value
  case 1
    print "Item 1 selected"
  case 2
    print "Item 2 selected"
end
```

SimTalk

Control [SimTalk] - DropDownList

See also

Select Object [for controls]

Configure the Check Box and the Drop-down List

Item [SimTalk] - DropDownList

Value [SimTalk] - DropDownList

Tab User-defined

Define your own attributes as described under the **Tab User-defined**.

Navigate Menu

The commands are described under the **Navigate Menu**.

View Menu

The commands are described under the **View Menu**.

SimTalk

updateDialog [SimTalk]

Tools Menu

The **Tools Menu** provides commands to access its functions.

Edit Controls

Edit Observers

See also

Tools Menu [general description]

Help Menu

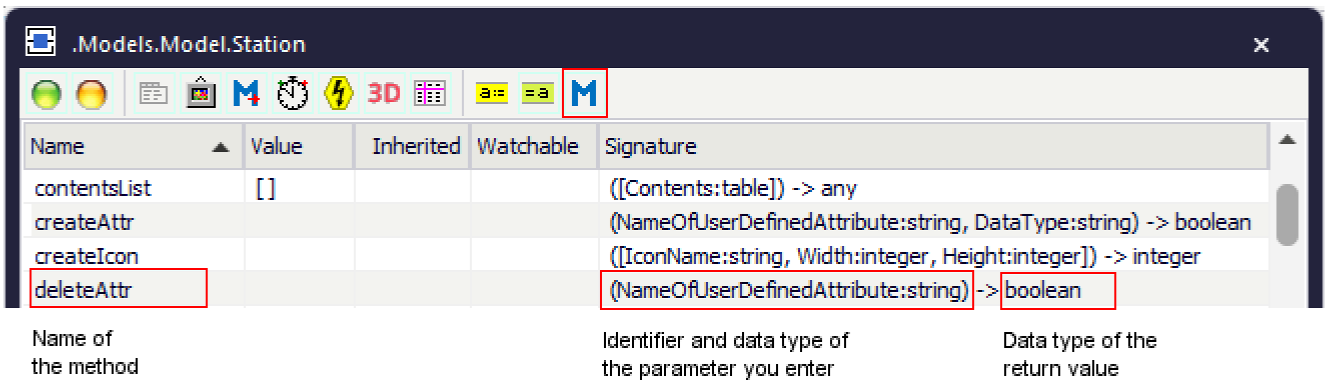
The commands are described under the **Help Menu**.

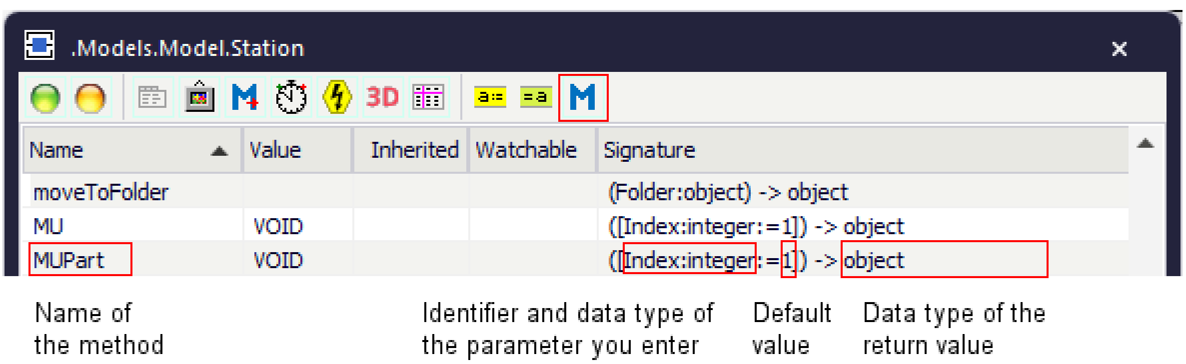
Methods of the DropDownList

Methods of the DropDownList

The *DropDownList* provides the **Methods of All Objects**.

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window **Show Attributes and Methods**. The figure below illustrates the information using the example of the object *Station*.





- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected **Class [general description]**.
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected **Instance [general description]**.

An example of the **Syntax** line of the individual methods might look like this:

```
<Path>.openDialog([CallOpenControl:boolean:=false]) → boolean
```